

Definiciones

• Z:

$$\begin{cases} V_1 = I_1 Z_{11} + I_2 Z_{12} \\ V_2 = I_1 Z_{21} + I_2 Z_{22} \end{cases}$$

$$Z_{\text{Top}} = \begin{bmatrix} Z_A + Z_B & Z_B \\ Z_B & Z_B + Z_C \end{bmatrix}$$

$$\bullet Z_{11} = \frac{V_1}{I_1} \Big|_{I_2=0}$$

$$\bullet Z_{12} = \frac{V_1}{I_2} \Big|_{I_1=0}$$

$$\bullet Z_{21} = \frac{V_2}{I_1} \Big|_{I_2=0}$$

$$\bullet Z_{22} = \frac{V_2}{I_2} \Big|_{I_1=0}$$

• Y:

$$\begin{cases} I_1 = V_1 Y_{11} + V_2 Y_{12} \\ I_2 = V_1 Y_{21} + V_2 Y_{22} \end{cases}$$

$$Y_{\pi} = \begin{pmatrix} Y_A + Y_B & -Y_B \\ -Y_B & Y_B + Y_C \end{pmatrix}$$

$$\bullet Y_{11} = \frac{I_1}{V_1} \Big|_{V_2=0}$$

$$\bullet Y_{12} = \frac{I_1}{V_2} \Big|_{V_1=0}$$

$$\bullet Y_{21} = \frac{I_2}{V_1} \Big|_{V_2=0}$$

$$\bullet Y_{22} = \frac{I_2}{V_2} \Big|_{V_1=0}$$

• T_{ABCD} :

$$\begin{cases} V_1 = V_2 A + (-I_2) B \\ I_1 = V_2 C + (-I_2) D \end{cases}$$

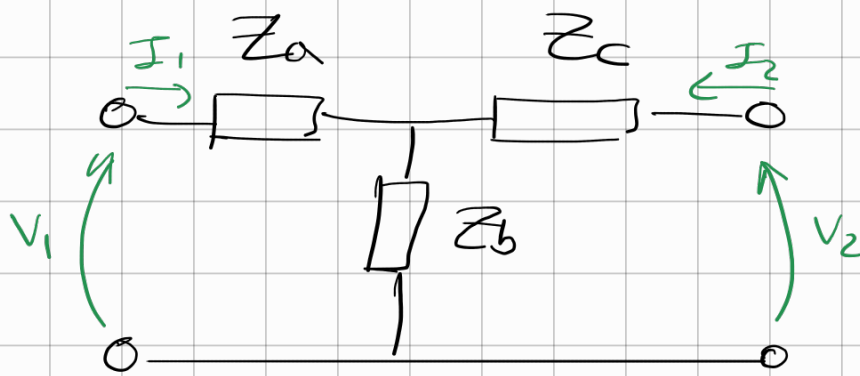
$$A = \frac{V_1}{V_2} \Big|_{I_2=0}$$

$$B = \frac{V_1}{-I_2} \Big|_{V_2=0}$$

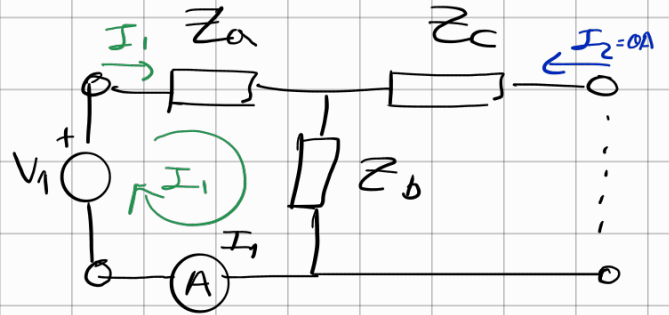
$$C = \frac{I_1}{V_2} \Big|_{I_2=0}$$

$$D = \frac{I_1}{(-I_2)} \Big|_{V_2=0}$$

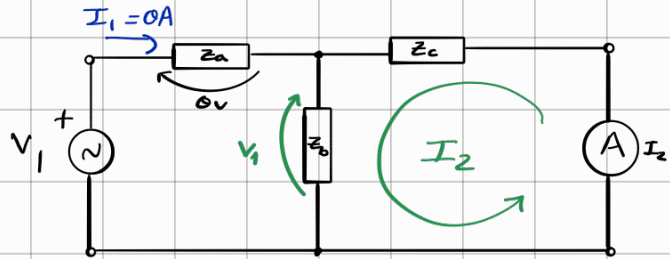
1. Tee)



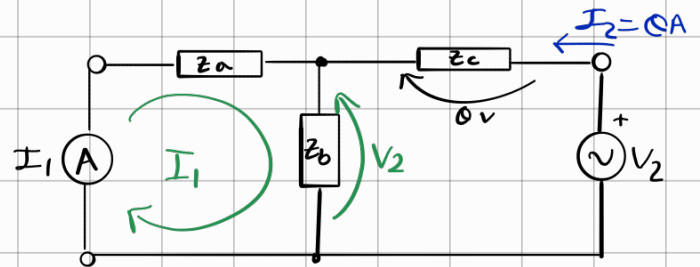
$$Z_{11} = \frac{V_1}{I_1} \bigg|_{I_2=0} = Z_a + Z_b //$$



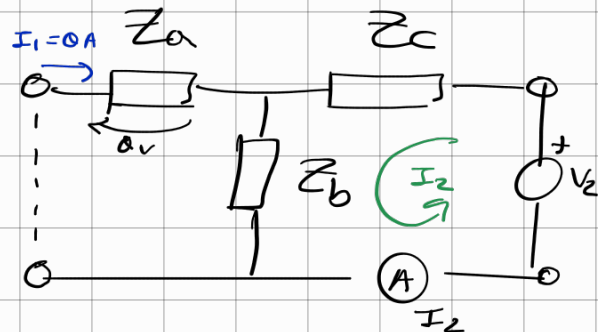
$$Z_{12} = \frac{V_1}{I_2} \bigg|_{I_1=0} = Z_b //$$



$$Z_{21} = \frac{V_2}{I_1} \bigg|_{I_2=0} = Z_b //$$



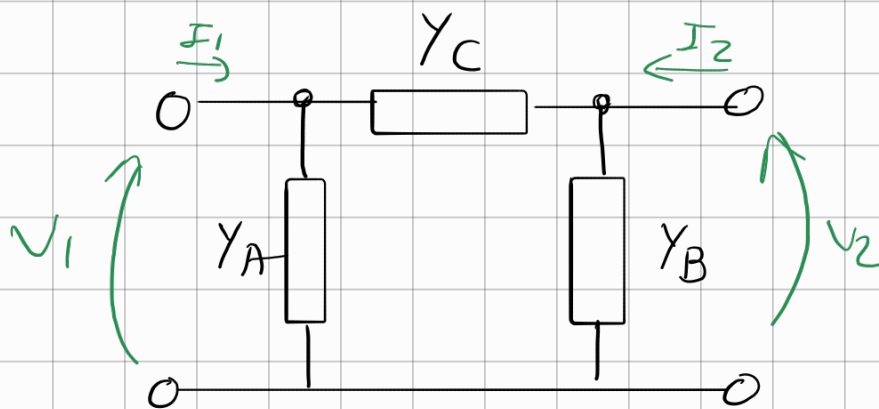
$$Z_{22} = \frac{V_2}{I_2} \bigg|_{I_1=0} = Z_b + Z_c //$$



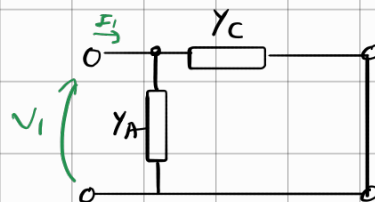
$$\underline{Z}_{Tee} = \begin{pmatrix} Z_a + Z_b & Z_b \\ Z_b & Z_b + Z_c \end{pmatrix}$$

$$\Delta Z = Z_a Z_b + Z_b Z_c + Z_c Z_a$$

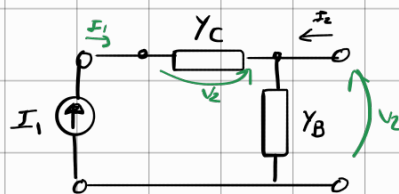
1. π)



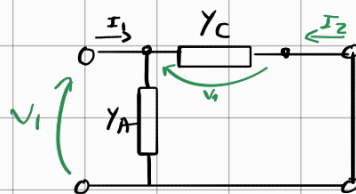
$$Y_{11} = \frac{I_1}{V_1} \Big|_{V_2=0} = Y_A + Y_C //$$



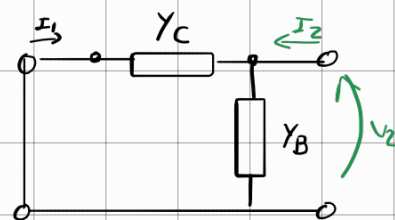
$$Y_{12} = \frac{I_1}{V_2} \Big|_{V_1=0} = -Y_C //$$



$$Y_{21} = \frac{I_2}{V_1} \Big|_{V_2=0} = -Y_C //$$



$$Y_{22} = \frac{I_2}{V_2} \Big|_{V_1=0} = Y_B + Y_C //$$

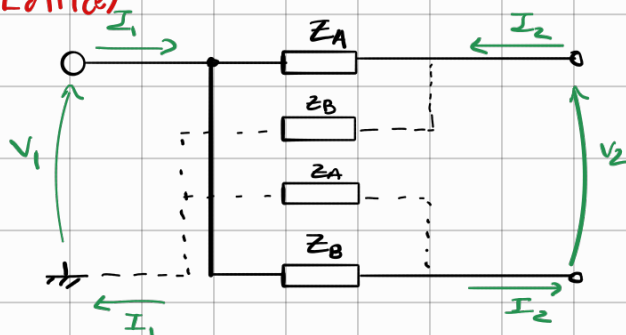


$$Y_{\pi} = \begin{pmatrix} Y_A + Y_B & -Y_B \\ -Y_B & Y_B + Y_C \end{pmatrix}$$

$$\Delta Y = Y_A Y_B + Y_B Y_C + Y_C Y_A$$

Lattice

1. Lattice)



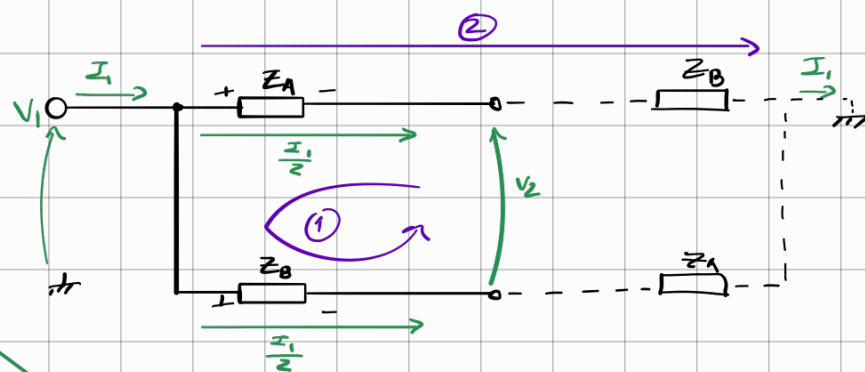
$$A = \frac{V_1}{V_2} \Big|_{I_2=0} \quad B = \frac{V_1}{(-I_2)} \Big|_{V_2=0}$$

$$C = \frac{I_1}{V_2} \Big|_{I_2=0} \quad D = \frac{I_1}{(-I_2)} \Big|_{V_2=0}$$

$I_2 = 0$

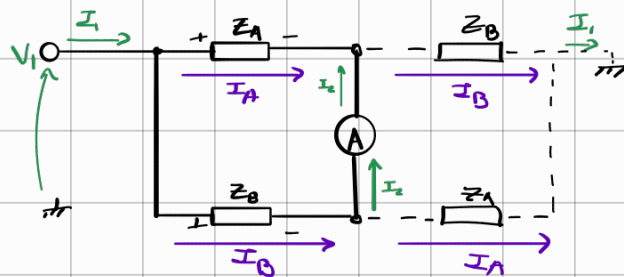
$$\begin{cases} \textcircled{1} V_2 = -Z_A \frac{I_1}{2} + Z_B \frac{I_1}{2} \\ \textcircled{2} V_1 = (Z_A + Z_B) \frac{I_1}{2} \end{cases}$$

$$A = \frac{V_1}{V_2} \Big|_{I_2=0} = \frac{Z_A + Z_B}{Z_B - Z_A}$$



$$C = \frac{I_1}{V_2} \Big|_{I_2=0} = \frac{2}{Z_B - Z_A}$$

$V_2 = 0$



$$Z_A I_A = Z_B I_B$$

$$I_1 = I_A + I_B$$

$$\begin{cases} I_A = I_1 \frac{Z_B}{Z_A + Z_B} \\ I_B = I_1 \frac{Z_A}{Z_A + Z_B} \end{cases}$$

$$I_B = I_2 + I_A$$

$$\Rightarrow I_1 \frac{Z_A - Z_B}{Z_A + Z_B} = I_2 \Rightarrow D = \frac{I_1}{-I_2} \Big|_{V_2=0} = \frac{Z_A + Z_B}{Z_B - Z_A}$$

$$V_1 = Z_A I_A + Z_B I_B = \frac{2Z_A Z_B}{Z_A + Z_B} \cdot \frac{Z_A + Z_B}{Z_B - Z_A} (-I_2) \Rightarrow B = \frac{V_1}{-I_2} \Big|_{V_2=0} = \frac{2Z_A Z_B}{Z_B - Z_A}$$

$$T_{ABCD} = \begin{pmatrix} \frac{Z_A + Z_B}{Z_B - Z_A} & \frac{2Z_A Z_B}{Z_B - Z_A} \\ \frac{2}{Z_B - Z_A} & \frac{Z_A + Z_B}{Z_B - Z_A} \end{pmatrix}$$

$$\Rightarrow \Delta T = 1$$

2. Tee)

$$\bullet Z \rightarrow Y: \begin{pmatrix} \frac{Z_{22}}{\Delta Z} & -\frac{Z_{12}}{\Delta Z} \\ -\frac{Z_{21}}{\Delta Z} & \frac{Z_{11}}{\Delta Z} \end{pmatrix} \Rightarrow$$

$$Y = \frac{1}{\Delta Z} \begin{pmatrix} Z_B + Z_C & -Z_B \\ -Z_B & Z_A + Z_B \end{pmatrix}$$

$$\Delta Z = Z_A Z_B + Z_B Z_C + Z_C Z_A$$

$$\bullet Z \rightarrow T: \begin{pmatrix} \frac{Z_{11}}{Z_{21}} & \frac{\Delta Z}{Z_{21}} \\ \frac{1}{Z_{21}} & \frac{Z_{22}}{Z_{21}} \end{pmatrix} \Rightarrow$$

$$T = \frac{1}{Z_B} \begin{pmatrix} Z_A + Z_B & \Delta Z \\ 1 & Z_B + Z_C \end{pmatrix}$$

$$\Delta Z = Z_A Z_B + Z_B Z_C + Z_C Z_A$$

2. Pi)

$$\bullet Y \rightarrow Z: \begin{pmatrix} \frac{Y_{22}}{\Delta Y} & -\frac{Y_{12}}{\Delta Y} \\ -\frac{Y_{21}}{\Delta Y} & \frac{Y_{11}}{\Delta Y} \end{pmatrix} \Rightarrow$$

$$Z = \frac{1}{\Delta Y} \begin{pmatrix} Y_B + Y_C & Y_B \\ Y_B & Y_A + Y_B \end{pmatrix}$$

$$\Delta Y = Y_A Y_B + Y_B Y_C + Y_C Y_A$$

$$\bullet Y \rightarrow T: \begin{pmatrix} -\frac{Y_{22}}{Y_{21}} & -\frac{1}{Y_{21}} \\ -\frac{\Delta Y}{Y_{21}} & -\frac{Y_{11}}{Y_{21}} \end{pmatrix} \Rightarrow$$

$$T = \frac{1}{Y_B} \begin{pmatrix} Y_B + Y_C & 1 \\ \Delta Y & Y_A + Y_B \end{pmatrix}$$

$$\Delta Y = Y_A Y_B + Y_B Y_C + Y_C Y_A$$

2. Lattice)

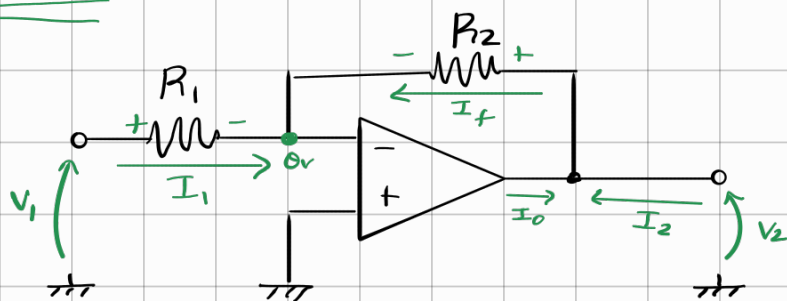
$$\bullet T \rightarrow Z: \begin{pmatrix} \frac{A}{C} & \frac{\Delta T}{C} \\ \frac{1}{C} & \frac{D}{C} \end{pmatrix} \Rightarrow$$

$$Z = \begin{pmatrix} \frac{Z_A + Z_B}{2} & \frac{Z_B - Z_A}{2} \\ \frac{Z_B - Z_A}{2} & \frac{Z_A + Z_B}{2} \end{pmatrix}$$

$$\bullet T \rightarrow Y: \begin{pmatrix} \frac{D}{B} & -\frac{\Delta T}{B} \\ -\frac{1}{B} & \frac{A}{B} \end{pmatrix} \Rightarrow$$

$$Y = \begin{pmatrix} \frac{Z_A + Z_B}{2Z_A Z_B} & \frac{Z_A - Z_B}{2Z_A Z_B} \\ \frac{Z_A - Z_B}{2Z_A Z_B} & \frac{Z_A + Z_B}{2Z_A Z_B} \end{pmatrix}$$

Bonus



$$\begin{cases} V_2 = -\frac{R_2}{R_1} V_1 \\ \hookrightarrow V_1 = -\frac{R_1}{R_2} V_2 // \\ I_1 = -I_F \\ I_2 = I_F - I_0 \end{cases}$$

○ $Z_{11} = \frac{V_1}{I_1} \bigg|_{I_2=0} \rightarrow V_1 = I_1 R_1 - I_F R_2 + V_2$

$$V_1 = I_1 (R_1 + R_2) - \frac{R_2}{R_1} V_1$$

$$\Rightarrow \boxed{Z_{11} = R_1}$$

○ $Z_{21} = \frac{V_2}{I_1} \bigg|_{I_2=0} \rightarrow V_1 = I_1 (R_1 + R_2) + V_2$

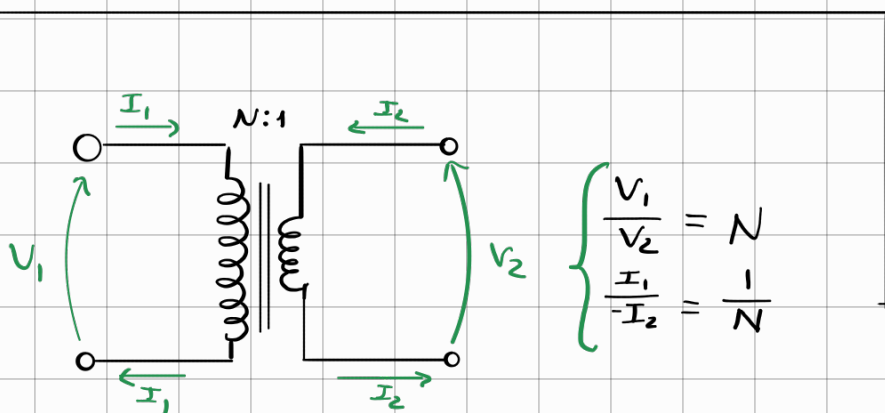
$$\left(1 + \frac{R_1}{R_2}\right) V_2 = -I_1 (R_1 + R_2)$$

$$\Rightarrow \boxed{Z_{21} = -R_2}$$

○ $Z_{22} = \frac{V_2}{I_2} \bigg|_{I_1=0} = Z_{OUT} \Rightarrow \boxed{Z_{22} = 0}$

○ $Z_{21} = \frac{V_1}{I_2} \bigg|_{I_1=0} = -\frac{R_1}{R_2} \cdot \frac{V_2}{I_2} \bigg|_{I_1=0} \Rightarrow \boxed{Z_{21} = 0}$

Debido a que la impedancia de salida del opam ideal es nula.



$$\begin{cases} \frac{V_1}{V_2} = N \\ \frac{I_1}{I_2} = \frac{1}{N} \end{cases}$$

$$\boxed{Z = \begin{pmatrix} R_1 & 0 \\ -R_2 & 0 \end{pmatrix}}$$

$$\Rightarrow \boxed{T_{ABCD} = \begin{pmatrix} N & 0 \\ 0 & 1/N \end{pmatrix}}$$